



CHEMISTRY Department
SIMHADRI

Note No :1

Ref: SMTTP/Chemistry/2024-25/OM CHEM/358091

Date:

29/07/2024(20:32:20)

Subject: The revision of LMI on SAFETY PRECAUTIONS FOR HANDLING BULK ACID CAUSTIC SODA AND SODIUM CHLORITE

The LMI/OGN/OPS/CHEM/007 on SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA AND SODIUM CHLORITE pertaining to O&M (CHEM) section has been reviewed and revised. Changes made are mentioned in the below table.

Para no	Revision details	Reason for Revision	Current Issue and revision details of LMI
	· Safety aspects for handling/unloading of hydrochloric acid/Caustic Soda/lye/ and Sodium Chlorite have been segregated into separate sections.	OGN Revised	LMI/OGN/OPS/CHEM/007
	· Separate Section for maintenance & Storage requirements of acid/alkali tanks /pumps have been introduced.	COS-ISO-00-OGN/OPS/CHEM/007 Rev. No.: 06, May 2024	ISSUE No: 08
	· SOP/JSA for rubber lining job of bulk acid storage tank have been incorporated in the document.		Date: 29-07-2024
	· Title of document revised from		

Safety Precautions for Handling
Bulk Acid and Caustic Soda

Annexure-
VIII.

- General Requirements for unloading acid/alkali by gravity been added
- Storage& Handling of Caustic soda flakes added

8.0

Submitted for kind approval please

Submitted by:

नागराजु भास्करानी - उप प्रबंधक (रसायन)

Nagaraju Bhaskarani - 103083 (DEPUTY MANAGER(CHEMIST) ,CHEMISTRY) ,Simhadri
(NAGARAJUBHASKARANI@NTPC.CO.IN, 9446004621)

On:29/07/2024(20:32:20) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:2

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:30/07/2024(18:50:33) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:3

Please review as discussed. Any deviation from OGN to be mentioned and justified in LMI and approval note.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:01/08/2024(18:07:31) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:4

As discussed please review.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:03/08/2024(10:40:48) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:5

Unloading and handling of Sulphuric Acid is being omitted as the station doesn't use the material.

Submitted by:

नागराजु भास्करानी - उप प्रबंधक (रसायन)

Nagaraju Bhaskarani - 103083 (DEPUTY MANAGER(CHEMIST) ,CHEMISTRY)
(NAGARAJUBHASKARANI@NTPC.CO.IN, 9446004621) ,Simhadri

On:03/08/2024(11:45:06) **Ref:**SMTTP/Chemistry/2024-25/OM CHEM/358091

Note No:6

LMI revised accordingly. Submitted for approval.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:03/08/2024(11:58:21) **Ref:**SMTTP/Chemistry/2024-25/OM CHEM/358091

Note No:7

Reviewed and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:07/08/2024(18:39:04) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:8

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:12/08/2024(11:31:20) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:9

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:14/08/2024(11:41:08) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:10

Revised as per standard format.

Submitted by:

नागराजु भास्करानी - उप प्रबंधक (रसायन)

Nagaraju Bhaskarani - 103083 (DEPUTY MANAGER(CHEMIST) ,CHEMISTRY)
(NAGARAJUBHASKARANI@NTPC.CO.IN, 9446004621) ,Simhadri

On:22/08/2024(16:35:13) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:11

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:22/08/2024(16:45:42) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:12

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:22/08/2024(17:17:38) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:13

Please review and comment

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)
(TVRSEKHAR@NTPC.CO.IN, 9650998176) ,Simhadri

On:22/08/2024(17:44:40) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:14

In LMI/OGN/OPS/CHEM/007, Activity Sl No-5 Sub clause-4.

colouring of pipeline Responsibility needs to be changed to CPC group. MMOS/chemistry may coordinate in identification of pipeline.

Submitted by:

हेमन्त सैनी - वरिष्ठ प्रबंधक

HEMANT SAINI - 100407 (SENIOR MANAGER ,MECH MAINT- OFF SITE)
(HEMANTSAINI@NTPC.CO.IN, 9425284675) ,Simhadri

On:30/08/2024(18:59:35) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:15

As per pre-page notings , MM OFS shall assist in the identification of pipelines. However, painting shall be done by CPC group.

Submitted please.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)
(TVRSEKHAR@NTPC.CO.IN, 9650998176) ,Simhadri

On:31/08/2024(11:24:49) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:16

Pl review the responsibility of C&I dept in this LMI

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:03/09/2024(17:00:05) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:17

Reviewed and found in order.

Submitted by:

शलिनी सिंह - उप महाप्रबंधक

Shalini Singh - 008621 (DY. GENERAL MANAGER ,C & I MAINT)
(SHALINISINGH01@NTPC.CO.IN, 9650991330) ,Simhadri

On:04/09/2024(17:56:25) **Ref:**SMTTP/Chemistry/2024-25/OM CHEM/358091

Note No:18

Forwarded for approval

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:04/09/2024(18:50:09) **Ref:**SMTTP/Chemistry/2024-25/OM CHEM/358091

Note No:19

Details of revisions to be added in the LMI

Submitted by:

सुरेश जॉन डेविड - महाप्रबंधक

Suresh John David - 005206 (GENERAL MANAGER ,O & M)
(SJDAVID@NTPC.CO.IN, 9447797541) ,Simhadri

On:10/09/2024(18:02:57) **Ref:**SMTTP/Chemistry/2024-25/OM CHEM/358091

Note No:20

May please add the detail of revisions in the LMI.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:14/09/2024(19:28:36) **Ref:**SMTTP/Chemistry/2024-25/OM CHEM/358091

Note No:21

Revision details added in the LMI

Submitted by:

नागराजु भास्करानी - उप प्रबंधक (रसायन)

Nagaraju Bhaskarani - 103083 (DEPUTY MANAGER(CHEMIST) ,CHEMISTRY)
(NAGARAJUBHASKARANI@NTPC.CO.IN, 9446004621) ,Simhadri

On:17/09/2024(18:40:59) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:22

Revision detail of the LMI is incorporated and copy is attached.

Submitted for approval please.

Submitted by:

सांतनु जेना - उप महाप्रबंधक (रसायन)

Santanu Jana - 007743 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(SANTANUJANA@NTPC.CO.IN, 7458012612) ,Simhadri

On:18/09/2024(11:35:59) **Ref:**SMTTPP/Chemistry/2024-25/OM CHEM/358091

Note No:23

May be approved

Submitted by:

सुरेश जॉन डेविड - महाप्रबंधक

Suresh John David - 005206 (GENERAL MANAGER ,O & M)
(SJDAVID@NTPC.CO.IN, 9447797541) ,Simhadri

On:18/09/2024(18:44:16) **Ref:**SMTPP/Chemistry/2024-25/OM CHEM/358091

Note No:24

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

Sanjay Kumar Sinha - 005185 (CGM ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:18/09/2024(18:48:35) **Ref:**SMTPP/Chemistry/2024-25/OM CHEM/358091

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**SAFETY PRECAUTIONS FOR HANDLING BULK ACID,
CAUSTIC SODA and SODIUM CHLORITE**

LMI/OGN/OPS/CHEM/007

JULY -2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
2	LMI No	LMI/OGN/OPS/CHEM/007
3	Reference Document	COS-ISO-00-OGN/OPS/CHEM/007, Rev. No.: 06 May 2024
4	Superseded Document	LMI/OGN/OPS/CHEM/007 Issue: 07, May 2022
5	Prepared By	Name: Nagaraju Bhaskaran, Dy. Manager (CHEM)
6	Checked By	Name: SANTANU JANA, DGM (CHEM)
7	Reviewed By	Name: S J David GM (O&M)
8	Approved By	Name: S.K.Sinha HOP (Simhadri)

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

SAFETY PRECAUTIONS FOR HANDLING BULK ACID AND CAUSTIC SODA AND SODIUM CHLORITE

INDEX

S. no	Description	Page Nos.
1.	Introduction	4
2.	Superseded Documents	4
3.	Revision details	4
4.	Scope	4
5.	SAFETY PRECAUTIONS TO BE TAKEN BEFORE UNLOADING OF HYDROCHLORIC ACID	4
6.	GUIDELINES FOR SAFE STORAGE/MAINTENANCE REQUIREMENTS FOR HYDROCHLORIC ACID TANKS AND PUMPS	7
7.	SAFETY PRECAUTIONS TO BE TAKEN BEFORE UNLOADING OF CAUSTIC SODA LYE	9
8.	GUIDELINES FOR SAFE STORAGE/MAINTENANCE REQUIREMENTS FOR CAUSTIC TANKS AND PUMPS	10
9.	STORAGE REQUIREMENT AND HANDLING OF CAUSTIC SODA FLAKES	13
10.	GUIDELINES FOR SAFE HANDLING/UNLOADING/STORAGE OF SODIUM CHLORITE	14
11.	EMERGENCY RESPONSE & HAZARD COMMUNICATION	15
12.	RECORDS TO BE MAINTAINED/ Reference Document/ Review	15
13.	REFERENCE DOCUMENT	15
14.	REVIEW	15
	Annexure – I Check list for unloading	16
	Annexure – II Check list for cleaning and repair	18
	Annexure – III MSDS Of Hydrochloric Acid	19
	Annexure- IV MSDS of Caustic Soda	24
	Annexure - V MSDS Of Sodium Chlorite Solution	29
	Annexure – VI Checklist for PPE's	33
	Annexure – VII GENERAL REQUIREMENTS FOR UNLOADING HYDROCHLORIC ACID/ ALKALI BY PUMPS	34
	Annexure – VIII GENERAL REQUIREMENTS FOR UNLOADING	35

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

	HYDROCHLORIC ACID/ ALKALI BY Gravity	
	Annexure – IX ANNUAL MAINTENANCE ACTIVITIES PROTOCOL FOR HYDROCHLORIC ACID SYSTEM	36
	Annexure – X MAINTENANCE ACTIVITIES PROTOCOL FOR CAUSTIC SYSTEM	38
	Annexure – XI SOP FOR RUBBER LINING IN BULK ACID TANKS	40
	Annexure – XII JSA FOR RUBBER LINING WORKS IN BULK ACID TANKS	42

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

1.0 INTRODUCTION

Hydrochloric acid and caustic soda are used in bulk quantities for the water treatment in power station. Some safety and administrative precautions need to be taken during receipt, unloading, storage and handling of bulk supplies of hydrochloric acid and caustic soda and also for the maintenance of bulk acid/alkali storage tanks.

2.0 SUPERSEDED DOCUMENTS

LMI/OGN/OPS/CHEM/007 Issue: 07, May 2022

3.0 Revision Details:

1. Safety aspects for handling/unloading of hydrochloric acid/Caustic Soda/lye/ and Sodium Chlorite have been segregated into separate sections.
2. Separate Section for maintenance & Storage requirements of acid/alkali tanks /pumps have been introduced.
3. **Annexure – XI** SOP/JSA for rubber lining job of bulk acid storage tank have been incorporated in the document.
4. **Annexure – VIII** General Requirements for unloading acid/alkali by gravity been added
5. Storage& Handling of Caustic soda flakes added.

4.0 SCOPE

This document establishes the requirements for the safety and administrative precautions, which need to be taken during receipt, unloading, storage and handling of bulk supplies of hydrochloric acid, caustic soda and sodium Chlorite. Unloading and handling of Sulphuric Acid is being omitted as the station doesn't use the material. It also describes the procedure to be adapted for the maintenance of bulk acid/alkali storage tanks.

5.0 SAFETY PRECAUTIONS TO BE TAKEN BEFORE UNLOADING OF HYDROCHLORIC ACID

S no	Activity	Responsibility
5.1		
1.	On receipt of Hydrochloric acid tanker, HOD (Chemistry) is to be intimated by concerned officer or Shift Chemist.	Shift Chemist
2.	Delivery note from the chemical tanker driver and results of station's laboratory testing should be used to identify the tanker contents before starting any unloading operation	-do-
3.	Every tanker should have NFPA DIAMOND displayed for the Chemical.	-do-
4.	The tanker should be rested for minimum half an hour before the commencement of Unloading activity	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

5.	ensure all following acid PPES compatible to Hydrochloric acid are available near unloading area(Annexure- VI to be maintained) A full-face shield / chemical safety goggles with side shield protection • Chemical protective suit • Chemical proof gloves • Gum boot • Gas mask with full face piece with acid gas canister and high efficiency particulate filter Resp: Shift Chemist	
6.	After the tanker is rested for half an hour, the tanker driver /his representative with all Chemical PPES should slowly open the manhole of the tanker and collect the samples	Shift Chemist
7.	During unloading of chemicals, unloading area should be cordoned off and unauthorized entry of personnel in the area is to be restricted. Caution signs must be placed at both the ends of the tanker to warn people approaching the tanker.	-do-
8.	The hose pipe connected to the tanker should be clamped properly. The same is to be ensured by Site MM-OFS representative.	MM(OFS)
9.	Check the operation of the Safety Shower & Eye wash fountain before unloading is started. Ensure water through safety shower & eye wash fountains is free of rust & dirt. The source of water used for safety Shower and eye washer should be from drinking water tapping.	Shift Chemist/Maint.
10.	Before connecting the Hose pipe for unloading, it is to be ensured that stoppers are provided at the tyres to restrict the tanker movement.	Shift Chemist
11.	Only one tanker at a time to be unloaded. In the event of a second tanker arriving at site while the first tanker is unloading, the second tanker must be parked away from unloading area.	-do-
12.	The movement of partially unloaded tankers should be avoided. However, if it is necessary to move a partially unloaded tanker, then make sure the tanker outlet valve is closed and all connections to the tanker are disconnected. The cover / vent of the tanker should be properly closed along with tightening of all the bolts.	-do-
13.	A hose pipe with running water shall be kept near the unloading point to take care of small leakages	-do-
14.	Emergency contact numbers and safety precautions/MSDS/first aid measures along with SOP for unloading in Hindi / English and local language should be displayed prominently in the unloading area.	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

15.	First aid in case of Contact with Acids: i. If acid gets into eyes, affected eyes are to be flushed thoroughly with water for 10-15 minutes take affected person to hospital. ii. If in contact with skin or clothing, remove clothing and wash skin thoroughly and take the affected person to hospital. iii. If swallowed, immediately wash mouth thoroughly with water. Report to hospital for immediate Medical Attention.	-do-
-----	---	------

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

S no	Activity	Responsibility
6.0	GUIDELINES FOR SAFE STORAGE/MAINTENANCE REQUIREMENTS FOR HYDROCHLORIC ACID TANKS AND PUMPS	
1.	Ensure the functioning of all level transmitters in storage tank.	HOD(C&I)/ Shift Chemist
2.	In case the level transmitter is not functioning, make sure to measure the level of storage tank by level indicator.	Shift Chemist
3.	Maximum storage capacity along with name of chemical should be mentioned on each tank. The date of last inspection done and the next due date for inspection should be displayed on the tank along with NPFA standard.	Shift Chemist
4.	The Color and labelling of the acid pipelines should be in accordance with the NTPC standard guidelines for colour coding scheme of pipelines.	HOD(Offiste MM)
5.	Preferably one storage tank should be kept in partially filled condition for emergency purpose	Shift Chemist
6.	Arrangements should be readily available for transferring the chemical from one tank to other tank in case of leakage from storage tank containing chemical. In case of storage constraints due to availability of 02 no's of permanent acid Storage tank, a temporary storage tank may be kept nearby for emptying the acid from leakage tank.	HOD (Offiste MM)/HOD Chem
7.	Storage tanks breathers'/fume absorbers should always be in working condition and inspection schedule is to be maintained during annual overhauling.	Shift Chemist
8.	Chemical storage area should have well ventilation facility with exhaust fans.	-do-
9.	Adequate water arrangements should be available for flushing any leak/ spill. All leaks /spills are to be attended on priority.	-do-
10.	Metallic ladder and platform both secured with handrails independent of the top of the tank should be available for every tank to reach its top for inspection, as it is unsafe to stand or walk on the tank top. Platform should also be available for safe operation of outlet valves of bulk chemical storage tank wherever approach is not available.	HOD(Offiste MM)
11.	Flange guards are to be provided by site maintenance personnel for all the flanges in the acid transfer line to prevent any accidental spray of chemical mist/fumes.	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

	For Hydrochloric Acid tank maintenance job, following activities are to be done and ensured before starting any maintenance activity:	
12	The storage tank should be filled with service water and to be drained, inlet and outlet lines disconnected and blanked off. Pipelines into or out of the tank should be disconnected, preferably by removing a complete small section. A blank flange should be secured on the open end of the line to protect against human error and unsuspected leaks. After this, it should be thoroughly purged with fresh air	HOD(Offiste MM)/HOD Chem
13	After ensuring complete draining and air purging, fill the storage tank first time with water solution and drain it. Continue filling and draining with service water till the effluent is free of acid.	Shift Chemist
14	Special ventilation and a continuous fresh air supply should always be available during the entire working time during cleaning, inspecting, or repairing of the storage tank.	Shift Chemist
15	Proper personal protective equipment for handling hydrochloric acid should be worn by everyone entering inside the storage tank for inspection, cleaning, or repairs.	-do-
16	One or two personnel should always stand outside the tank should keep constant observation on the men working inside the tank and at least two other men should be available nearby to help those in the tank in case of trouble.	OFS/Chem
17	A hose mask, an air-line mask, or a self-contained breathing apparatus, together with rescue harness and lifeline, should be located outside the tank entrance. This equipment should be used for rescue, if necessary.	-do-
18	Only portable electric light of 24-volt DC supply should be used inside the storage tank.	-do-
19	Annual overhauling of both bulk acid storage/measuring tanks should be done once in a year . A record for the same is to be maintained by Site representatives of Chemistry, MM-OFS & FQA. Before any internal inspection of Acid tanks, specified safety precautions should be taken along with PTW/Confined issued by concerned person. Oxygen level and flammable gases level to be checked before the entry of any person inside the tank. Safety belt should be used for opening the manhole of Storage tanks. Annexure-IX, II to be maintained.	-do-
20	A detailed JSA for the rubber lining job for bulk acid storage tank is to be prepared by site in accordance with their scope of work and Job plan. All associated safety hazards like fire and chemical hazard should be covered in the JSA. An illustrative JSA is given in Annexure-XII for ready reference.	-do-
21	Time bound internal rubber lining replacement of the of Storage and measuring tank in case of degradation of rubber quality as per standard guidelines. Rubber lining job to be completed within 30 days	HOD(Offiste MM)

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

	after issuance of PTW.	
22	Replacement of manual valves, flanges and gaskets associated with tanks for safe operation and handling of acids.	HOD(Offiste MM)
23	Replacement of temporary hose pipe (used for unloading) at every two years should be done. It is recommended to use Rubber Braided hose or UHMWPE hose. Hose integrity to be checked once in a month including end connections and protocol for the same is to be maintained by site maintenance representative.	-do-
24	Pipelines associated with tanks is to be inspected and to be replaced in case of any abnormality.	-do-
25	Maintenance of Acid Unloading pumps /dosing pumps to be carried out as per PM schedule. If the PM Schedule is not followed, exceptions for the same may be recorded. Flange/gaskets for both acid unloading/dosing pumps header also to be replaced annually.	-do-
26	Availability /Calibration of Pressure gauges/transmitter in the acid dosing/unloading pumps should be ensured for safe operation of pumps.	HOD(C&I)
27	Site is to keep a minimum of 20% spares quantity of all the Valves/flanges/gaskets associated with acid system for catering to the emergency in addition to the planned maintenance activity	HOD (Offiste MM)
28	A SIGNED PROTOCOL BY MM-OFS/CHEMISTRY IS TO BE MAINTAINED FOR THE LIST OF ACTIVITIES CARRIED OUT DURING THE MAINTENANCE OF THE STORAGE TANK- Annexure-IX	OFS/Shift Chem

7.0	SAFETY PRECAUTIONS TO BE TAKEN BEFORE UNLOADING OF CAUSTIC SODA LYE	Responsibility
1	On receipt of Caustic Soda tanker, HOD (Chemistry) is to be intimated by concerned officer/Shift Chemist. Delivery note from the chemical tanker driver and results of station's laboratory testing should be used to identify the tanker contents before starting any unloading operation. The tanker should have NFPA diamond displayed.	Shift chemist
2	The tanker should be rested for minimum half an hour before the commencement of Unloading activity.	Shift Chemist
3	Site to ensure all following acid PPES compatible to caustic are available near unloading area: • A full-face shield / chemical safety goggles with side shield protection • Chemical protective suit • Chemical proof gloves • Gum boot	-do-
4	After the tanker is rested for half an hour, the tanker driver /his representative with all Chemical PPES should slowly open the manhole of the tanker and collect the samples.	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

5	During unloading of chemicals, unloading area should be cordoned off and unauthorized entry of personnel in the area is to be restricted. Caution signs must be placed at both the ends of the tanker to warn people approaching the tanker.	-do-
6	The hose pipe connected to the tanker should be clamped properly. The same is to be ensured by Site MM-OFS representative. Before connecting the Hose pipe for unloading, it is to be ensured that stoppers are provided at the tyres to restrict the tanker movement. Ensure water through safety shower & eye wash fountains is free of rust & dirt. The source of water used for safety Shower and eye washer should be from drinking water tapping.	Shift Chemist
7	Only one tanker at a time to be unloaded. In the event of a second tanker arriving at site while the first tanker is unloading, the second tanker must be parked away from unloading area.	-do-
8	The movement of partially unloaded tankers should be avoided. However, if it is necessary to move a partially unloaded tanker, then make sure the tanker outlet valve is closed and all connections to the tanker are disconnected.	-do-
9	A hose pipe with running water shall be kept near the unloading point to take care of small leakages	-do-
10	Checklist as per Annexure – 1 must be filled up before start of each unloading process.	-do-
11	Emergency contact numbers and safety precautions/MSDS/first aid measures along with SOP for unloading in Hindi / English and local language should be displayed prominently in the unloading area	-do-
12	First aid in case of Contact with alkali: a. If alkali gets into eyes, affected eyes are to be flushed thoroughly with water for 10-15 minutes, take affected person to hospital. b. If in contact with skin or clothing, remove clothing and wash skin thoroughly, take affected person to hospital. c. If swallowed, immediately wash mouth thoroughly with water. Report to Hospital for immediate Medical Attention.	
8.0	GUIDELINES FOR SAFE STORAGE/MAINTENANCE REQUIREMENTS FOR CAUSTIC TANKS AND PUMPS	Responsibility
1.	Ensure the functioning of all level transmitters in storage tank. In case the level transmitter is not functioning, make sure to measure the level of storage tank by level indicator.	Shift Chemist
2.	a. Maximum storage capacity along with name of chemical should be mentioned in each tank. The date of last inspection done and the next due date for inspection should be clearly displayed on the tank along with NPFA standard. The Color and labelling of the	Shift Chemist

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

	alkali pipelines should be in accordance with the NTPC standard guidelines for color coding scheme of pipelines. Preferably one storage tank should be kept in partially filled condition for emergency purpose. Arrangements should be readily available for transferring the chemical from one tank to other tank in case of leakage from storage tank containing chemical. In case of storage constraints due to availability of 02 nos of Storage tank, a temporary storage tank may be kept nearby for emptying the alkali from leakage tank.	
3	Chemical storage area should be well ventilated with exhaust fans	HOD(Chemistry)
4	Adequate water arrangements should be available for flushing any leak/ spill.	Shift Chemist
5	Any alkali leakage in the pipelines or tank body should be attended on priority.	HOD(Offsite Maint)
6	Metallic ladder and platform both secured with handrails independent of the top of the tank should be available for every tank to reach its top for inspection, as it is unsafe to stand or walk on the tank top. Platform should also be available for safe operation of outlet valves of bulk chemical storage tank wherever approach is not available.	HOD(Offsite Maint)
8.2	For Caustic lye tank maintenance job, following activities are to be done and ensured before starting any maintenance activity:	
1.	The storage tank should be filled with service water and to be drained, inlet and outlet lines disconnected and blanked off. Pipelines into or out of the tank should be disconnected, preferably by removing a complete small section. A blank flange should be secured on the open end of the line to protect against human error and unsuspected leaks. After this, it should be thoroughly purged with air.	Shift Chemist/MM oFS
2.	After ensuring complete draining and air purging, fill the storage tank first time with water solution and drain it. After complete draining, continue filling and draining with service water till the pH of the draining solution is 7.5-8.5.	Shift Chemist
3.	Special ventilation and a continuous fresh air supply should always be available during the entire working time during cleaning, inspecting, or repairing of the storage tank.	-do-
4.	Proper personal protective equipment for handling caustic lye should be worn by everyone entering inside the storage tank for inspection, cleaning, or repairs	Shift Chemist/MM OFS

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

S no	Activity	Responsibility
5	One or two personnel should always stand outside the tank should keep constant observation on the men working inside the tank and at least two other men should be available nearby to help those in the tank in case of trouble.	HOD(offsite MM)
6	Only portable electric light of 24-volt DC supply should be used inside the storage tank..	HOD(offsite MM) &HOD (Chemistry)
7	Overhauling of both caustic storage/measuring tanks should be carried out once in two years. A record for the same is to be maintained by Site representatives of Chemistry, MMOFS & FQA. Before any internal inspection of Alkali tanks, specified safety precautions should be taken along with PTW issued by concerned person. Safety belt should be used for opening the manhole of Storage tanks. Annexure-X,II to be maintained.	HOD(offsite MM)/Shift Chemist
8	SOP for rubber lining job and detailed JSA is to be prepared by site maintenance offsite personnel as per the scope of work and job planned. EIC should ensure that all persons working for rubber lining job should wear PPES during the job and should not be allowed to carry any inflammable substances with them.	HOD(offsite MM) &HOD (Chemistry)
9	Following jobs are to be mandatorily carried during the overhauling:	
10	Time bound internal rubber lining replacement of the of Storage and measuring tank in case of degradation of rubber quality as per standard guidelines. Rubber lining job to be completed within 30 days after issuance of PTW	- HOD(offsite MM)
11	Replacement of manual valves, flanges and gaskets associated with tanks for safe operation and handling of alkali.	-do-
12	Pipelines associated with tanks is to be inspected and to be replaced in case of any abnormality	-do-
13	Maintenance of Alkali Unloading pumps /dosing pumps to be carried out as per PM schedule. If the PM Schedule is not followed, exceptions for the same may be recorded. Flange/gaskets for both alkali unloading/dosing pumps header also to be replaced annually.	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

14	Availability /Calibration of Pressure gauges/transmitter in the alkali dosing/unloading pumps should be ensured for safe operation of pumps.	HOD (C&I)
15	A SIGNED PROTOCOL BY MM-OFS/CHEMISTRY IS TO BE MAINTAINED FOR THE LIST OF ACTIVITIES CARRIED OUT DURING THE MAINTENANCE OF THE STORAGE TANK. (Refer Annexure X). Site is to keep a minimum of 20% spares quantity of all the Valves/flanges/gaskets associated with alkali system for catering to the emergency in addition to the planned maintenance activity	HOD(offsite MM) &HOD (Chemistry)
9.0	STORAGE REQUIREMENT AND HANDLING OF CAUSTIC SODA FLAKES	Shift Chemist
1.	Generally caustic soda flakes are supplied in 50 kgs HDPE bag with HDPE liner and stored at a designated place either in Central store or, in Water treatment plant area.	Shift Chemist
2.	Due to hygroscopic nature, Caustic soda bags should be stored in dry, cool and well-ventilated area, preferably on an elevated platform such as on PVC pallets or, any other acid/alkali resistant materials like rubber, PVC etc. Caustic soda bags should be stored in covered shed / room, away from direct sun light and rain, separated from in compatible materials.	-do-
3.	Hooks should not be used for loading / unloading of caustic soda bags	-do-
4.	SOLUTION PREPARATION OF CAUSTIC FLAKES	
5.	Mild steel rubber lined / SS tanks with basket shall only be used for caustic soda solution preparation. These tanks should have stirrer and recirculation facility	-do-
6.	While preparing caustic solution from flakes, NEVER ADD WATER TO CAUSTIC SODA FLAKES. Always add flakes to water with constant stirring to avoid a violent exothermic reaction. Use cold water to prevent excessive heat generation.	-do-
7.	Take D.M. Water in the caustic preparation tank up to the bottom level of the basket so that at least 20 to 30 cm water remains in the basket. Start the stirrer and add caustic soda flakes in the basket slowly with constant stirring. (If sodium hydroxide becomes concentrated in one area or, if added too rapidly; a rapid temperature increase can result in dangerous boiling and/or spattering or may cause an immediate	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

	violent eruption.)	
8.	While preparing the solution, it is to be remembered that a lot of heat is generated during dissolution of caustic soda flakes in water, thus making the caustic soda preparation tank very hot	-do-
9.	After all the requisite quantity of caustic soda flakes has been added to the preparation tank, the caustic solution must be recirculated for at least one to two hours so to have a uniform caustic lye.	-do-
10	The contents of damaged bags must be promptly used or must be placed in suitable airtight Container.	-do-
11	After preparation of the caustic solution, it can be transferred and stored in caustic storage tanks or caustic day tanks, which should be made up of MSRL or SS.	-do-
10.0	GUIDELINES FOR SAFE HANDLING/UNLOADING/STORAGE OF SODIUM CHLORITE	Shift Chemist
1.	In view of the use of Sodium Chlorite (25-31%) used in the plant for generation of Chlorine Dioxide, it is recommended to take the following general measures while Handling, Storage and Neutralization of Sodium Chlorite.	Shift Chemist
2.	During Unloading, all necessary safety precautions to be taken. Protective measure for handling: i. avoids contact with skin, eyes, membranes, and clothes. ii. use neoprene gloves and boots, goggles, and fire-proof coverall. iii. keep the mask at hand in case of possible ClO ₂ evolutions. iv. keep rooms ventilate	-do-
3.	Protective measure for storage: i. Keep away from acids, combustible substances and heat sources. ii. Select a fresh and ventilated place. iii. Materials to avoid steel, copper and its alloys, aluminum, natural and synthetic rubber. iv. Suitable materials: glass, PVC, polyethylene, polyester, and steel (AISI316 or similar)	-do-
4.	General guidelines for Neutralization of Sodium Chlorite NaClO₂ in ClO₂ plant.	-do-

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

5.	Due to the potential for heat generation, it is important to dilute the sodium chlorite with water to a solution containing no more than 5% (wt.) sodium chlorite. When diluting sodium chlorite in pit, the pit should never be filled beyond 75% of its capacity	-do-
6.	Slowly and carefully, add dry sodium sulfite OR SHMBS. The addition of an excess of sodium sulfite /SHMBS will assure the complete reduction of chlorite and will prevent the generation of chlorine dioxide in the next step. As a precaution, sodium sulfite should be added slowly to allow for adequate heat dissipation.	-do-
7.	Once the chlorite has been completely reacted, slowly add a 5% hydrochloric acid (HCl) (prepared in chemical preparation tank) solution to lower the pH to 4-5.	Shift Chemist
11.0	EMERGENCY RESPONSE & HAZARD COMMUNICATION:	Responsibility
1	If proper Safety measures are practiced no emergency conditions will arise during Handling of Acids/ Caustic soda and Sodium Chlorite. If any emergency condition arises HOD (Chemistry) and HOD (Safety) shall be intimated, and all activities should be carried out with their directions to control the emergency situation. Actions should be taken as per Disaster Management Plan of station. Annexed standards for Material Safety Data Sheets (MSDS) should be referred for further details	-do-
12.0	RECORDS TO BE MAINTAINED:	Shift Chemist
	Record description 1. Weekly Availability of PPES as per Annexure -VI 2. Acid/alkali unloading Checklist as per Annexure-1 3. Maintenance records of Bulk acid/alkali system including storage tank inspection records As per Annexure II,VII,VIII,IX,X. 4. SOP /JSA for rubber lining job as per annexure, XI, XII	Maintained by 1. Shift Chemist 2. Shift Chemist 3. Shift Chemist /MM-OFS 4. MM-OFS

13.0 REFERENCE DOCUMENT:

COS-ISO-00-OGN/OPS/CHEM/007 Rev. No.: 06 May 2024

14.0 REVIEW

HOD (CHEM) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

ANNEXURE –I**O&M Chemistry:** Checklist for receiving and unloading of Acid/Caustic Lye

NAME OF CHEMICAL:

Date:-

ARRIVAL TIME:

UNLOADING START TIME:

Sl. No.	Description	Status	Remarks if any
1	Access of tanker up to unloading point is safe.	Yes / No	
2	Tanker has been weighed in store.	Yes / No	
3	Test certificate checked and the identity of the chemical established.	Yes / No	
4	TREM card from the driver is checked.	Yes / No	
5	Availability of Safety and eye shower with clean water is checked.	Yes / No	
6	Tanker's manhole is opened	Yes / No	
6	Running water hose is available near unloading point.	Yes / No	
7	Hose pipe clamping to tanker ensured by Site MM-OFS	Yes / No	
8	Proper storage tank to receive the material is identified.	Yes / No	
9	The storage tank has sufficient capacity to receive the material.	Yes / No	
10	Inlet V/V of other tanks closed.	Yes / No	
11	Before starting unloading, the tanker's vent valve is opened fully.	Yes / No	
12	Ensure stopper provide at tyre to restrict the tanker's movement.	Yes / No	
13	Tanker driver and helper are wearing the necessary PPES before the start of unloading activity	Yes / No	
14	Required PPE's are available – Full face shield - Chemical resistant apron - Chemical proof gloves - Gum boot	Yes / No Yes / No Yes / No Yes / No	
15	Laboratory test report confirms the quality as per P.O.	Yes / No	
16	Initial and final tank level		
17	Any other Issues/Remarks		
Signature			

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

IS STANDARDS :

S.NO	NAME OF CHEMICAL	REFERENCE STANDARD
<u>1</u>	HYDROCHLORIC ACID	6164- 1967.latest version
<u>2</u>	CAUSTIC SODA /LYE	4264- 1967.latest version
<u>3</u>	SULPHURIC ACID	4262- 1967/& BIS 266.latest version

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

ANNEXURE – II

Checklist for Acid/Alkali storage tank inspection or Repair

Name of Tank:

Date:

SL. No.	Description	Status	Remark
1	Ensure permit is issued for inspection or maintenance	Yes / No	
2	Ensure identity of tank to be inspected	Yes / No	
3	Ensure inlet valve is closed and red tag is put	Yes / No	
4	Ensure vessel is cleaned at least two times with water so no acid/alkali fume is coming out	Yes / No	
5	Ensure Drain valve is opened	Yes / No	
6	Ensure Manhole(s) kept in opened condition	Yes / No	
7	Ensure Fresh air circulation, tank, with service air pipe line	Yes / No	
8	Ensure sufficient lighting in tank (only portable 24volt DC light is allowed)	Yes / No	
9	Ensure availability of hose pipe with running water	Yes / No	
10	Ensure availability of safety shower and eye shower	Yes / No	
11	Ensure a SCBA together with harness located outside the tank entrance	Yes / No	
12	Ensure one man outside of tank to keep watch on men working inside and Ensure at least two other men nearby to help in case of any trouble	Yes / No	
13	Ensure Person to enter vessel should use PPEs	Yes / No	
	Gas mask		
	Chemical resistant apron		
	Chemical proof gloves		
14	Ensure minimum oxygen level of 19.5 % inside the tank	Yes / No	
15	Ensure work at height safety	Yes / No	
	Proper ropes and anchors	Yes / No	
	Helmet	Yes / No	
	Safety belt	Yes / No	

Chemistry dept
Signature:
Name:

MM- Dept
Signature:
Name:

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

ANNEXURE- III

MATERIAL SAFETY DATA SHEET: HYDROCHLORIC ACID

Section 1:- COMPOSITION / INGREDIENTS

Hydrochloric Acid % :- 32 –35 %

Chemical Formula: - HCl

Molecular Weight :- 36.46 gm/mole

CAS No :- 7647-01-0

TLV :- 5 PPM (7mg/M₃) for 8 hours working

Section 2 :- HAZARD & TOXICITY

Danger! Hydrochloric Acid and its fumes Cause eye and skin burns. Causes digestive and respiratory tract burns. May be fatal if inhaled or swallowed.

The following effects are observed on exposure to different concentrations of hydrogen chloride in air:

30 ppm by volume -Causes irritation of throat

50 to 100 ppm by volume- Tolerated for not more than an hour

1000 to 2000 ppm by volume- Dangerous even for brief exposure

Local

a) Eyes - Contact of the eyes with hydrochloric acid vapour or mist rapidly causes severe irritation of the eyes and eyelids. If the acid is not quickly removed by thorough irrigation with water, there may be prolonged or permanent visual impairment or total loss of eye-sight. Inflammation of the eye is characterized by redness, watering, and itching.

b) Skin - Concentrated solutions on contact with the skin, cause severe burns unless promptly washed off. Repeated contact of the skin with dilute solutions may lead to the development of dermatitis.

Systemic

a) Inhalation - Inhalation of excessive concentrations of hydrochloric acid vapours immediately produces severe irritation of the respiratory tract with sore throat, coughing, shortness of breath. Such an exposure necessitates prompt voluntary withdrawal from the contaminated atmosphere. Causes chemical burn to the respiratory tract. Causes corrosive action on the mucous membranes. Under rare conditions, if inhaled deeply, edema of the lungs may occur.

b) Ingestion - Causes severe digestive tract burns with abdominal pain, vomiting, and possible death. May cause corrosion and permanent tissue destruction of the oesophagus and digestive tract. The lips and mouth usually turn white, and later, brown.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Section 3 :- FIRST AID MEASURES

General - Remove the affected person from the contaminated atmosphere and removing hydrochloric acid from the skin or eyes is of primary importance. First-aid shall be started immediately in all cases of contact with hydrochloric acid in any form.

Contact with Skin - In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes as rapidly as possible while they are in the shower. **No attempt should be made to neutralize the acid with alkaline solutions.** Get medical aid immediately.

Contact with Eyes – Immediately flush eyes with copious quantities of running water for at least 15 minutes. The eyelids should be held apart during the irrigation to ensure contact of water with all accessible tissues of the eyes and rotating the eye balls. DO NOT allow victim to rub eyes or keep eyes closed. Get medical aid immediately. If a physician is not immediately available, the eye irrigation should be continued for a second period of 15 minutes.

Inhalation - If inhaled, move to fresh air. Loosen tight clothing such as a collar, tie, belt or waistband. If breathing is difficult, give oxygen. Oxygen usually furnishes relief from coughing. Give artificial respiration ONLY if breathing has stopped. Do not use mouth-to-mouth method if victim ingested or inhaled the substance: Induce artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention immediately.

Ingestion - If swallowed, do NOT induce vomiting. If spontaneous vomiting occurs, have victim lean forward with head down to avoid breathing in of vomitus, rinse mouth and administer more water. Get medical aid immediately. If victim is fully conscious, he should immediately be made to drink copious amounts of lime water or milk of magnesia. Use plain water if these are not readily available. Never give anything by mouth to an unconscious person.

Section 4: - FIRE FIGHTING MEASURES

Hydrochloric acid, as such, involves no fire hazard, but reacts with most metals to form EXPLOSIVE hydrogen gas. Use water spray to keep fire-exposed containers cool. Vapors may be heavier than air. They can spread along the ground and collect in low or confined areas. As in any fire, wear a self-contained breathing apparatus in pressure-demand.

Section 5: - ACCIDENTAL RELEASE MEASURES

Spilled hydrochloric acid should always be attended to. Move people away from the area of spill. Move upwind. Avoid contact with acid. Isolate area and deny entry. Prevent material from entering confined spaces, sewers or waterways.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Small Spill: Dilute with water and mop up, or absorb with an inert dry material and place in an appropriate waste disposal container. Soda ash or lime should be available for immediately neutralizing spilled acid on concrete, wood, or other reactive or absorbing material. When soda ash is used, ample ventilation should be provided.

Large Spill: Stop leak if without risk. Absorb with DRY earth, sand or other non-combustible material. Use water spray curtain to divert vapour drift. Use water spray to reduce vapours. Be careful that the product is not present at a concentration level above TLV. Contact Fire Brigade for disposal.

Section 6:- HANDLING & STORAGE

General – For large volume, Hydrochloric Acid should be stored in Mild Steel Rubber Lining Tanks (MSRL). Hydrochloric acid storage should be in the open, or in well-ventilated buildings or sheds. Ample, natural ventilation should be provided. All tanks are to be located within a dike area.

Type of Construction - Since concrete surfaces disintegrate in contact with hydrochloric acid, acid-proof brick is the preferred material of construction. Steel, concrete and wood, when used, should be protected by suitable acid-resistant surface coatings. Provision should be made for drainage and washing down spills with large quantities of water.

Isolation- Hydrochloric acid should not be stored with or near oxidizing agents, particularly nitric acid and chlorates. Fumes from hydrochloric acid are highly corrosive to metals and thus, indirectly, present health and fire hazards.

Venting of Tanks- Since hydrochloric acid fumes settle rather rapidly toward the ground, it is recommended that the fumes be carried through acid-resistant ducts to a centrally located scrubbing tower / Fume Absorber. Water can be used in the tower to absorb the acid fumes for conveying it to a neutralizing pit. An acid-resistant overflow, should be provided either in combination with, or separate from the venting system and should discharge into tile drain or sump located well away from building or tank foundations.

Handling – Wear suitable protective clothing, gloves and eye / face protection while handling hydrochloric acid. In case of insufficient ventilation, wear suitable respiratory equipment. Change and wash clothing and personal protective equipment if contaminated before storing and /or reusing. Wash hands and face thoroughly after handling and before work breaks, eating, drinking and using toilet facilities.

Section 7:- EXPOSURE CONTROLS & PERSONAL PROTECTION

General - Hydrochloric acid does not present a serious industrial hazard, provided workers are properly instructed and adequately supervised in handling it.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

The most important factor in the prevention of injury is to prevent contact of the gas or solutions with the eyes, skin, and the respiratory and gastrointestinal tracts.

Engineering Controls: Provide exhaust ventilation or other engineering controls to keep the airborne concentrations of vapours below their respective threshold limit value of 5 ppm by volume of air. Ensure that eyewash stations and safety showers are provided at the work-station location. The showers should be capable of supplying large quantities of water under moderately high pressure. Special eye-washing fountains, a bubbler drinking fountain, or a hose with a liberal, gentle flow of drinking water should be available for eye irrigation. All of the above equipments should be inspected and tested at frequent intervals to ensure its being in good working condition at all times. The locations of such equipment should be marked clearly.

Employee Education - Employees working where hydrochloric acid hazards exist should be instructed and trained in the following:

- a) Hazards of hydrochloric acid.
- b) Location of gas masks and other personal protective equipment, safety showers, bubbler drinking fountains, equipment especially suitable for eye irrigation, water hoses, exits, and first-aid equipment.
- c) Proper use of personal protective, emergency, and first-aid equipment.
- d) Proper individual conduct in case of emergency.
- e) The necessity for reporting to his superior any possible hazard, such as an unusual odour of hydrogen chloride or any evidence of spillage / leakage of hydrochloric acid.
- f) Key personnel should be drilled in the proper use of emergency equipment and in the actions to be taken in case of acid burns

Personal Protective Equipment - *Personal protective equipment is precautionary measures to avoid accidents and not a substitute for unsafe working conditions.* Employees who handle hydrochloric acid should be provided with the following equipment:

- **Eyes:** Splash proof and face shield goggles
- **Hands:** Protective neoprene or, polyethylene gloves
- **Clothing:** Disposable latex/ rubber apron, rubber boots with pant legs over boots.
- **Respirators:** A self-contained breathing apparatus, or an air-line mask
*Industrial gas masks equipped with canisters can be used only when it is certain that the concentration of vapour is less than 2 percent by volume (20,000 parts per million) and the oxygen content is not less than 16 percent, and then only for exposure **not exceeding half an hour.** This type of mask should not be used in an emergency situation.*

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Section 8:- PHYSICAL & CHEMICAL PROPERTIES.

STATE: -	Liquid
APPEARANCE: -	Colourless to Pale Yellow
ODOUR: -	Pungent, Irritating (Strong.)
PH: -	Strongly Acidic
BOILING POINT: -	83 °C @ 760 mmHg
FREEZING POINT: -	Minus 66 °C
SPECIFIC GRAVITY: -	1.150 to 1.164
VAPOUR PRESSURE: -	84 mm Hg @ 20 °C(35%)
VAPOUR DENSITY: -	1.27 (air=1)
CORROSIVITY: -	Highly corrosive to most metals with evolution of explosive hydrogen gas.
HYGROSCOPICITY: -	Hygroscopic
Solubility in Water: -	Soluble in all proportions.
Solubility in Other Liquids: -	Very soluble in ethanol, methanol, dioxane and tetrahydrofuran; insoluble in hydrocarbons, e.g. n-hexane.

Section 9:- STABILITY & REACTIVITY

STABILITY: - Stable under normal temperatures and pressures.

Conditions to Avoid: - Excess heat.

Incompatibilities with Other Materials:

Corrosive to most metals and their alloys, concrete, some plastics and all most all bases. Mixing with strong acids can cause evolution of hydrogen chloride gas. Oxidising agents will cause the release of toxic chlorine gas. Contact of acid with alkali or active metal may develop enough heat to cause fire in adjacent combustible material.

Hazardous Decomposition Products:-

Hydrogen chloride, chlorine, hydrogen gas.

REFERENCE:- IS: 6164- 1971, Reaffirmed 2003

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

ANNEXURE-IV**MATERIAL SAFETY DATA SHEET: CAUSTIC SODA LYE****Section 1 :- COMPOSITION / INGREDIENTS**

Sodium Hydroxide % :- 48 %

Chemical Formula :- NaOH

Molecular Weight :- 40.0 gm/mole

CAS No :- 1310-73-2

TLV :- 2 mg/M₃**Section 2 :- HAZARDS IDENTIFICATION*****EMERGENCY OVERVIEW***

Clear, colourless, odourless, dense, oily liquid. Extremely corrosive! Contact can cause eye burns and permanent tissue damage. May be fatal if swallowed. Signs and symptoms of irritation do not immediately become evident after contact with caustic soda; as a result, the worker may suffer injury before he realizes that the chemical is in contact with his body.

Acute Toxicity

Local:- Caustic soda destroys living tissues. Considerable heat is generated on adding water to caustic lye and this may cause boiling and spattering of hot solution.

a) Eyes:- Eye contact of only a few seconds can cause permanent damage, even blindness.

b) Skin:- Contact may cause severe irritation or corrosive skin damage, depending upon the length of contact, amount of caustic and concentration.

Systemic

a) Inhalation - Inhalation of the caustic aerosols including mists, vapours, fog and other airborne forms of any particle size may cause damage to upper respiratory tract and even to the lung tissue proper. The effects of irritation may vary from mild irritation of nasal membranes to severe pneumonitis depending on the severity of the exposure. Causes chemical burns to the respiratory tract.

b) Ingestion - Ingestion may cause severe caustic burns to the mouth, throat, oesophagus and stomach. Gross ingestion may cause death. May cause corrosion and permanent tissue destruction of the oesophagus and digestive tract.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Section 3 :- FIRST AID MEASURES

General - Speed in removing the affected person from the contaminated atmosphere and in removing Caustic Soda from the affected part is of primary importance. First-aid shall be started immediately in all cases of contact with Caustic Soda in any form.

Contact with Skin - In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes as rapidly as possible while they are in the shower. If irritation persists, repeat flushing. Do not transport victim unless the recommended flushing period is completed or flushing can be continued during transport. Get medical aid immediately.

Contact with Eyes – Immediately flush eyes with copious quantities of running water for at least 15 minutes. The eyelids should be held apart during the irrigation to ensure contact of water with all accessible tissues of the eyes and rotating the eye balls. DO NOT allow victim to rub eyes or keep eyes closed. Get medical aid immediately. If a physician is not immediately available, the eye irrigation should be continued for a second period of 15 minutes.

Inhalation - If inhaled, remove to fresh air. Loosen tight clothing such as a collar, tie, belt or waistband. If breathing is difficult, give oxygen. Oxygen usually furnishes relief from coughing. Give artificial respiration ONLY if breathing has stopped. Do not use mouth-to-mouth method if victim ingested or inhaled the substance: induce artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical aid immediately.

Ingestion - If swallowed, do NOT induce vomiting. If spontaneous vomiting occurs, have victim lean forward with head down to avoid breathing in of vomitus. Rinse mouth and administer more water. Get medical aid immediately. If victim is alert and not convulsing, rinse mouth and give 1 to 2 glass of water to dilute material. Never give anything by mouth to an unconscious person.

Section 4: - FIRE FIGHTING MEASURES

Sodium Hydroxide is non-flammable and non-combustible. However, during a fire, irritating / toxic oxides of Na_2O may be generated.

As in any fire, wear a self-contained breathing apparatus in pressure-demand. Approach fire from upwind to avoid hazardous vapours and toxic decomposition products.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Suitable Extinguishing Media: Carbon dioxide. Foam. Dry chemical power. Use flooding quantities of water as fog or, spray to keep fire exposed containers cool. Avoid direct water use, if possible.

Section 5: - ACCIDENTAL RELEASE MEASURES

Spilled Caustic Lye should always be attended to. Move people away from the area of spill. Move upwind. Avoid contact with the chemical. Isolate area and deny entry. Prevent material from entering confined spaces, sewers or waterways.

Small Spill: Soak up spill with absorbent material that does not react with spilled chemical. Put material in suitable covered, labeled containers. Flush area with water.

Large Spill: Stop leak if without risk. Completely contain spilled material with dikes, sandbags, etc., and prevent run-off into ground or surface waters or sewers. Recover as much material as possible into containers for disposal. Remaining material may be diluted with water and neutralized with dilute hydrochloric acid. Neutralization products, both liquid and solid, must be recovered for disposal.

Section 6:- HANDLING & STORAGE

General – Caustic Lye storage should be in the open, or in well-ventilated buildings or sheds. Ample, natural ventilation should be provided. All tanks are to be located within a dike area.

Type of Construction - For large volume, Caustic Lye should be stored in Mild Steel Rubber Lining Tanks (MSRL) or Stainless Steel Tanks.

Venting of Tanks- Each storage tank should be provided with a vent of sufficient size to maintain the tank at atmospheric pressure. An overflow, should be provided either in combination with, or separate from, the venting system and should discharge into tile drain or sump located well away from building or tank foundations.

Floors of the storage area should be of acid resistant material and should have adequate slope.

Isolation- Caustic Lye should not be stored with or near strong acids, organic halogen compounds, nitro compounds and amphoteric metals, such as Aluminium, Magnesium And Zinc.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Handling – This material is a CORROSIVE and VERY TOXIC liquid. Wear suitable protective clothing, gloves and eye/face protection while handling. In case of insufficient ventilation, wear suitable respiratory equipment. Change and wash clothing and personal protective equipment if contaminated before storing and/or reusing. Wash hands and face thoroughly after handling and before work breaks, eating, drinking and using toilet facilities.

Section 7:- EXPOSURE CONTROLS & PERSONAL PROTECTION

General - Personal protection scheme depends upon the effectiveness of employee's education, training and supervision. *Personal protective equipment is precautionary measures to avoid accidents and not a substitute for unsafe working conditions.*

Engineering Controls: Ensure that eyewash stations and safety showers are provided at the work-station location. The showers should be capable of supplying large quantities of water under moderately high pressure. Special eye-washing fountains, a bubbler drinking fountain, or a hose with a liberal gentle flow of drinking water should be available for eye irrigation. All of the above equipments should be inspected and tested at frequent intervals to ensure its being in good working condition at all times. The locations of such equipment should be marked clearly.

Employee Education - Employees required to handle caustic soda in any form should be instructed about the hazards involved and safe methods of work. They should be provided with adequate and appropriate personal protective equipment and its correct use should be ensured. They should be instructed about basic first-aid measures against alkali and the steps to be taken in the event of spillage of caustic soda. They should be instructed about the location and use of showers, eye fountains, hose lines etc. Food should not be stored or eaten in a work-area where caustic soda is handled.

Personal Protective Equipment - Employees handling Caustic Lye should be provided with the following equipment:

Eyes: Splash proof and face shield goggles.

Hands: Protective neoprene or, polyethylene gloves.

Clothing: Disposable latex/ rubber apron, rubber boots with pant legs over boots.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Section 8 :- PHYSICAL & CHEMICAL PROPERTIES.

STATE: -	Liquid
APPEARANCE: -	Colourless
ODOUR: -	Odourless
pH :-	Strongly Alkaline
BOILING POINT :-	145 °C @ 760 mmHg
FREEZING POINT :-	12.0 °C
SPECIFIC GRAVITY :-	1.5 to 1.53
VAPOUR PRESSURE :-	6.3 mm Hg at 40°C
CORROSIVITY :-	Highly corrosive
HYGROSCOPICITY :-	Highly Hygroscopic
Solubility in Water :-	Soluble in all proportions with generation of heat.
Solubility in Other Liquids: -	Very soluble in ethanol, methanol

Section 9:- STABILITY & REACTIVITY

STABILITY :- Stable under normal temperatures and pressures.

Conditions to Avoid: Mixing with water, acid or incompatible materials may cause splattering and release of large amounts of heat. Will react with some metals forming flammable hydrogen gas.

Incompatibilities with Other Materials: Acids, Halogenated compounds, Prolonged contact with aluminium, brass, bronze, copper, lead, tin, zinc or other alkali sensitive metals or alloys.

Hazardous Decomposition Products: Toxic fumes of Sodium oxide.

Hazardous Polymerization: Will not occur.

REFERENCE:- IS: 4264- 1967, Reaffirmed 2002

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

ANNEXURE-V
MATERIAL SAFETY DATA SHEET

SODIUM CHLORITE SOLUTION

1. PRODUCT IDENTIFICATION

Chemical abstract name: sodium chlorite

CAS number: 7758-19-2

EINECS number: 231-836-6

Formula: NaClO₂

Molecular weight: 90.436

2. COMPOSITION - INGREDIENTS

Hazardous Components % Classification R Phrases 25 - 31 % sodium chlorite 100 Xn (harmful) R 22 (harmful if swallowed) in aqueous solution

R 32 (contact with acids liberates very toxic gas)

R 41 (risk of serious damage to eyes)

3. HAZARD DATA

Health: the product is harmful if swallowed and may cause damage to eyes.

Environment: when in contact with acids, the product liberates chlorine dioxide - very toxic and explosive gas. Do not scatter in the environment.

4. FIRST AID

Eye contact: wash immediately with plenty of water for at least 15 minutes. Seek medical advise.

Skin contact: take off immediately all contaminated clothing. Wash immediately the affected part with water and neutral soap.

If swallowed: do not induce vomiting. Give plenty of water to drink. Seek immediately medical advice.

Inhalation: immediately move subject to fresh air. Keep patient warm and quiet. In case of inhalation of chlorine dioxide, if breathing is difficult, give oxygen and seek immediately medical advice. Never give anything orally if the victim is unconscious and do not give any kind of antidotes unless expressly prescribed.

5. FIRE HAZARD DATA

Extinguishing media: water is the most suitable extinguishing medium.

Sodium chlorite is not flammable but, in any case, if the product is involved in fire, cool

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

containers exposed to fire with sprinkled water. If clothes are burning, use plenty of water

rather than covers to extinguish them.

Extinguishing media to avoid CO₂ Eventual risk of exposure: O₂, acid fumes and Na₂O fumes
 Personnel equipment: self-contained breathing apparatus.

6. ACCIDENTAL SPILL RESPONSE

Remove incompatible substances. Circumscribe the concerned area to prevent the product from entering drains/streams. Alert local authorities. Remove spilled fluid with sand, earth or other inert absorbent material taking appropriate individual measures. Collect the product in emergency containers. Clean up residues with plenty of water and neutralize with reducing agents (sodium sulfite/bisulfite).

7. HANDLING AND STORAGE

Protective measure for handling:

- avoid contact with skin, eyes, membranes and clothes
- use neoprene gloves and boots, goggles and fire-proof overalls
- keep the mask at hand in case of possible ClO₂ evolutions
- keep rooms ventilated. If necessary, install air suction equipment in work places.

CHLORINE DIOXIDE

Protective measure for storage:

- keep away from acids, combustible substances and heat sources.

Select a fresh and ventilated place.

- materials to avoid: steel, copper and its alloys, aluminum, natural and synthetic rubber
- suitable materials: glass, PVC, polyethylene, polyester and steel (AISI 316 or similar)

8. INDUSTRIAL HYGIENE/PERSONAL PROTECTIVE EQUIPMENT

Sanitary inspections: in compliance with applicable national and local regulations

Exposure limits: TLV-TWA (ACGIH 1994/5) - 0.1ppm (as chlorine dioxide)

TLV-STEL (ACGIH 1994/5) - 0.3ppm (as chlorine dioxide)

Respiratory protection: mask with toxic gas filter

Hands protection: neoprene gloves

Eyes protection: goggles

Skin protection: appropriate protective clothing (fire-proof overalls, neoprene boots)

Technical protection of system: room ventilation, local air suction

9. PHYSICAL AND CHEMICAL DATA

Physical state: aqueous solution

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Color: straw yellow

Odor: odorless

pH: approximately 12

Flash point: not flammable

Freezing point: -3 °C (25 % type); 3 °C (31 % type)

Boiling point: decomposes at $t^{\circ} > 140^{\circ}\text{C}$ (decomposition products: chlorate, chloride, can also liberate oxygen)

Combustion hazard: the dried solution is a strong oxidizing agent

Density: 1.21 - 1.22 (25 % type); 1.26 - 1.28 (31 % type) g/ml at 15°C

Solubility in water: complete (at 20 °C it is 400 g/l of NaClO₂)

Solubility in solvents: insoluble

Viscosity at 20 °C: mPa.s: 2.33 (25 % solution); 3.26 (31 % solution)

Organic content: NIL

10. STABILITY AND REACTIVITY

Conditions to avoid- Avoid heating product and its consequent concentrations.

Materials to avoid: The solution does not pose any fire or explosion risk (as it is) Any spills must be removed before the solution dries, because it can generate flammable and explosive mixtures with combustible substances and easily oxidizable materials such as wood, pulp, oils, cotton. Contact with acids liberates chlorine dioxide, very toxic and explosive gas. Hazardous decomposition products: O₂, acid fumes and Na₂O fumes in case of fire.

11. TOXICOLOGIC INFORMATION

Ways of penetration: ingestion, inhalation, contact

Acute toxicity: LD₅₀ = 532 mg/Kg (rat) for 31 % type chlorite

LD₅₀ = 660 mg/Kg (rat) for 25 % type chlorite

Corrosiveness/irritating power:

- skin: irritant
- eyes: very irritant
- respiratory system: in case of accidental formation of aerosols you may have:
- risk of severe irritation of respiratory tract
- risk of methaemoglobinemia

These risks have been based on animal experimentation.

12. ECOLOGICAL INFORMATION

Possible effects, behavior and transformation in the environment: use in compliance with good work practices, avoiding to scatter the product in the environment.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

13. RECOMMENDED DISPOSAL

Safety/handling of waste: re-use if possible. Handle in accordance with applicable local and

national regulations.

Disposal methods:

- Waste: authorized chemical/physical treatment plants
- Product containers: reclaim by rinsing with water

Note: the user must be sure to observe other applicable national and local regulations.

14. TRANSPORTATION INFORMATION

Classification: UN N°: 1908 - road and railway transport:

RID/ADR: Class 8 Ord. 61° b) Label N° 8 Identification N°: 80/1908 -carriage by sea:

IMDG Code: Class 8-page 8137 (Amdt 27-94) Label N° 8 + Marine Pollutant EmS 8-06 MFAG 741 - air transportation: ICAO/IATA: Class 8 - packing class: II MONO

15. REGULATION INFORMATION

Classification and labelling: interim

Symbol of danger: Xn

Indication of danger: harmful

Nature of special risks: R 22: harmful if swallowed

R 32: contact with acids liberates very toxic gas

R 41: risk of serious damage to eyes

Safety advice: S 14: keep away from acids

S 26: in case of contact with eyes, rinse immediately with plenty of water and seek medical advice

S 36/37/39: wear suitable protective clothing, gloves and eyes/face protection

16. ADDITIONAL INFORMATION

Use of sodium chlorite: to produce ClO₂ in the disinfection of water destined for human consumption, waste waters, industrial processes, deodorization treatments, textile leaching and pulp bleaching.

The information supplied in this MATERIAL SAFETY DATA SHEET which represents the total sum of our knowledge and experience with the product is, by no means, exhaustive. It concerns the product as it is and in accordance with specifications. In case of combinations and mixes, make sure that hazards are not revealed. The user shall always be sure to meet any local regulations concerning the product, industrial hygiene and safety in the workplace.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

ANNEXURE-VI

Weekly Checklist for Availability of PPE'S

Location:

Date:

PPE	LOCATION	TOTAL NO'S AVAILABLE	WORKING	REMARKS IF ANY
Full-face shield				
Chemical safety goggles with side shield protection				
chemical protective suit.				
Chemical proof gloves				
Gum Boot				
Gas mask with full face piece with acid gas canister and high efficiency particulate filter.				EXPIRY DATE TO BE MENTIONED
Self-contained breathing apparatus with a full face piece operated in pressure demand mode or other positive pressure mode				
Any others				

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Annexure-VII.**GENERAL REQUIREMENTS FOR UNLOADING HYDROCHLORIC ACID/ ALKALI BY PUMPS**

For Hydrochloric acid, ensure adequate quantity of Lime/ Soda Ash is available in nearby area of unloading and storage area for Acid unloading area to meet any exigency. The unloading platform should have a ramp wide enough to rest front wheels of the tanker to provide a downward slope to the rear end to the tanker so that all the chemicals can be extracted from the tanker.

Select the storage tank to which the chemical will be pumped into and ensure that it has sufficient capacity to receive delivery of the quantity to be transferred

Open the storage tank's inlet valve in which delivery of acid is to be taken and ensure that inlet valves of all other storage tanks are closed. Check and record the initial level of the storage tank before unloading.

Select which unloading pump is to be used. (It will be better to ensure that both un-loading pumps are available and switched on from the MCC Room)

After ensuring the Quality (as specified in P.O.), connect the flexible hose with tankers outlet valve (Flexible hose connection is to be provided by Mechanical maintenance personnel) and open the valve slowly. If any leak observed, close the valve and attend the leakage. Then open the tanker vent valve slowly and ensure that the vent valve is opened to prevent any implosion of tank during unloading.

Open the common suction header valve, pump suction valve and air release valve of the unloading pump to ensure that suction line is filled with chemical, and no air is entrapped.

Close air release valve and start the unloading pump, discharge valve to be opened slowly to maintain required discharge pressure of around 1-2kg/cm².

Run the pump till complete unloading by throttling the discharge valve of the pump when the pressure drops so that complete quantity is unloaded. Stop the unloading pump and close all the valves.

Disconnect the flexible hose from the tanker side carefully. Flush the unloading pump and suction line & flexible hose with water. After unloading process is complete, pipe for the unloading pump is to be kept in dummy condition to prevent any contaminant ingress.

Record the final level and the level rise of the filled storage tank.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Annexure-VIII.**GENERAL REQUIREMENTS FOR UNLOADING HYDROCHLORIC ACID/CAUSTIC LYE BY GRAVITY:**

This can only be done where suitable facility (like ramp of desired height) is not available or permanent unloading facility is not available.

Collect sample from tanker for testing quality of the unloading acid/ alkali. Concentration and other quality parameters should be as per purchase order conditions.

Select the storage tank to which the chemical will be transferred and ensure that it has sufficient capacity to receive delivery of the quantity to be transferred. Record its initial tank level.

Connect the delivery valve of tanker with flexible hose to the common main header of storage tanks (Flexible hose connection is to be provided by Mechanical maintenance personnel) and then the vent valve of the tanker is to be opened fully.

Open that storage tank's inlet valve in which delivery of acid/ alkali is to be taken, also ensure that inlet valves of all other storage tanks are closed. Check and record the initial level of the storage tank before unloading.

After ensuring the Quality (as specified in P.O.), open the delivery valve of the tanker. Let the process to continue till all material of the tanker is transferred to storage tank and no further level rise is observed.

Record final level and level rise of the filled storage tank.

NOTE: If any leaks observed during the delivery, the transfer should be stopped immediately. Tanker delivery valves shut and area to be washed down before dealing with leaks. In event of any leakage/ spillage during the chemical handling or unloading process, HOD (Chemistry) and HOD (Safety) should be intimated immediately. For safe Chemical Unloading, the total activity is to be done in presence of Site MM-OFS representative.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Annexure-IX**ANNUAL MAINTENANCE ACTIVITIES PROTOCOL FOR HYDROCHLORIC ACID SYSTEM:**

Equipment Details:	Equipment (Storage tank/measuring tank):	Capacity:	Location:
Last OH/ inspection date:			
PTW issue date			
Date of start of Maintenance:			
PTW closing and return date.			
LIST OF JOBS/ACTIVITIES			
Activities	STATUS OF JOBS		Remarks
Inspection of rubber lining inside the tank carried out .	YES	NO	
Spark test done. (min 4KV per 1mm thickness)	YES	NO	Details may be mentioned. Patch work may be done for localized defect
Hardness – Shore A scale testing done	YES	NO	For Fresh lining – 60±5 Consider for re lining if more than 40% of the readings exceeds 85 or 20% exceeds 90.
Whether total rubber lining replacement is done.	YES	NO	Patch repair may be done for smaller areas
Storage tank fume chamber /breather inspection done.	YES	NO	
Any maintenance jobs done in the fume chamber.	YES	NO	
Inspection of pipelines associated with tank.	YES	NO	
Replacement of Storage tank outlet valves.	YES	NO	
Replacement of manual valves, flanges and gaskets associated with the tanks done.	YES	NO	

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE		
LMI No.	LMI/OGN/OPS/CHEM/007		

Maintenance of Acid Unloading /dosing pumps done as per PM schedule.	YES	NO	
Inspection of pipelines associated with Acid unloading pumps/dosing pumps.	YES	NO	
Replacement of Valves, flanges associated with Acid dosing and Unloading system	YES	NO	
Replacement of unloading hose pipe done.	YES	NO	
Any other Jobs:	YES	NO	

(MM-OFFSITE)

(CHEMISTRY)

(FQA)

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Annexure-X - MAINTENANCE ACTIVITIES PROTOCOL FOR CAUSTIC SYSTEM

Equipment Details:	Equipment (Storage tank/measuring tank):	Capacity:	Location:
Last OH/ inspection date:			
PTW issue date:			
Date of start of Maintenance:			
PTW closing and return date.			
LIST OF JOBS/ACTIVITIES			
Activities	STATUS OF JOBS		Remarks:
Inspection of rubber lining inside the tank carried out.	YES	NO	
Spark test done. (min 4KV per 1mm thickness)	YES	NO	Details may be mentioned. Patch work may be done for localized defect
Hardness – Shore A scale testing done	YES	NO	For Fresh lining – 60±5 Consider for re lining if more than 40% of the readings exceeds 85 or 20% exceeds 90
Whether total rubber lining replacement is done.	YES	NO	Patch repair may be done for smaller areas
Inspection of pipelines associated with tank.	YES	NO	
Replacement of Storage tank outlet valves.	YES	NO	
Replacement of manual valves, flanges and gaskets associated with the tanks don	YES	NO	Details may be specified
Maintenance of Alkali	YES	NO	Details may be

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE		
LMI No.	LMI/OGN/OPS/CHEM/007		

Unloading /dosing pumps done as per PM schedule.			specified
Inspection of pipelines associated with Alkali unloading pumps/dosing pumps	YES	NO	Details of the maintenance jobs may be mentioned
Replacement of Valves, flanges associated with Alkali dosing and Unloading system	YES	NO	Details to be specified.
Replacement of unloading hose pipe done.			
Any other Jobs:			

(MM-OFFSITE)

(CHEMISTRY)

(FQA)

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Annexure-XI- SOP for Rubber lining in Bulk Acid Storage Tanks:

Objective: The objective of this SOP is to ensure that all activities are carried out safely during rubber lining of Bulk Acid Storage tank. The scope covers for patch work and complete rubber lining.

Maintenance of Rubber-lined Vessels & Tanks If there is emergency when any corrosion on rubberlined vessels and tanks is noticed, immediately isolate the unit, open, take out the media or charge, dismantle the internals and check the continuity of rubber-lining by spark test. Wherever defect noticed, repair the same with patch work if possible.

Patch work to be done by cutting rubber in the affected area ensuring at least six inch more length is cut around the area, then repairing the affected metal surface ensuring at least six inch more length is cut around the area, then repairing the affected metal surface ensuring wet Hessian sack are kept on lining all around while cutting or welding, surface cleaning, putting new lining and curing applying hot electric iron, testing the final lining for continuity and hardness. It is preferable expert is taken for patch work.

Even if there is no emergency, check the lining once in a year as a routine and take necessary steps. Note: The Storage tanks must be filled with water up to 75% of the capacity if acid is not available for longer durations to prevent rubber lining damage.

The criteria for re rubber lining of a complete tank is if it fails during spark testing at many places and if rubber is found in damaged condition, you should go for re lining. The Quality Plan must be followed for the rubber to be used in rubber lining as per IS 3400 Part 1 & lining in accordance with IS 4682 Part 1. Procedure to be adopted for Rubber lining of Acid.

Storage tank:

1. Identify the criticality of the work to be done for both patch and complete lining inside the acid storage tank.
2. The tank must be filled with water and drained after transfer of acid.
3. Prior to Acid Storage tank yearly inspection for quality checks, make sure the tank is adequately ventilated. Two no: Manholes must be made available for all the Acid & Alkali storage tanks. Exhaust Fans must be used.
4. Confined space permit, Work at height permit must be taken in addition to Isolation PTW.
5. Supervision of the activity should be done one of the persons all the while during the Activity of lining removal and relining.
6. Removal of damaged rubber lining. Burning of old rubber is not allowed.
7. Surface Preparation of metal surfaces to be done by blasting

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

8. Application of Adhesive on the metal solution
9. Take necessary precautions as the adhesive solution used for rubber lining is highly inflammable. Instructions to the workers must be given daily about the hazards of Rubber lining process.
10. Pasting of rubber sheets. Ensure proper bonding of sheets
Proceed for Vulcanization of rubber by steam ingress inside the tank to achieve the desired hardness of rubber. Steam of 120°C - 160°C may be used of max. Pressure 4bar.
Insulation/Covering of Vessel during vulcanization is recommended.
12. No need for steam ingress if already vulcanized sheets are used for rubber lining.
13. Proceed for Quality checks after 48hrs of completion of vulcanization.
14. In case of welding/patch welding for storage tanks, proper care must be taken.
15. All the activities must conform to IS 4682 Part 1
16. Adhesion Test may be done for every rubber lot on test specimen.
17. Spark test for the rubber lining should be done with a minimum voltage of 4-5KV/mm of rubber lining.

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Annexure-XII- JSA for Rubber lining in Bulk Acid Storage Tanks

JOB SAFETY ANALYSIS	JSA Number:	JSA/O&M/MMOFS /TANK_RUBBER LINING/001	Date:	
	JSA Done by:			
Work Instruction /Job Plan				
JOB (Brief description)	RUBBER LINING WORKS INSIDE CHEMICAL STORAGE VESSELS IN DM PLANT/CPU PLANT			
JOB Location			Dept.	Offsite Mntc
Job Plan / Step	Hazards	Control Measures	Residual Risk (High/Medium/Low)	Person responsible for Control Measure
Confined space work • Accessing Height location	Working inside vessel. • Working at Height	Ensure Confined space PTW • Ensuring Work at height permit, • Use of Double harness safety belt, rigid anchorage points	HIGH	
Ensuring there are no fumes/ strong odour of the chemical inside the tank	Confined space, darkness, lack of ventilation, Toxic gases/vapours	Oxygen and harmful gases to be measured by competent person. Proper ventilation to be ensured. • All Manholes in the tank are to be opened and cordoned off	HIGH	
Provide access ladder	Slip/fall while climbing down into	Ensure 3-point contact while climbing down the ladder, provide	MEDIUM	

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

	the vessel	guide rope, Ensure the access ladder is properly tied to prevent slip		
Internal inspection	Suffocation inside duct	Proper ventilation and forced air to be ensured. • Provide 24 V lighting inside the vessel, Ensure enough illumination	MEDIUM	
Removal of old/ damaged rubber on the interior surface	Cut during rubber removal.	Provide 24 V lighting inside the vessel, Ensure enough illumination	LOW	
Surface preparation by copper slag blasting, Grinding	Noise during blasting Hit by Object	Workers to come out of the vessel for fresh air period at regular periods, ensure usage of PPEs- face shield, nose mask, Hood to protect face, neck, shoulders during blasting. h. • Hydro test and thickness measurement to be done periodically for the vessel (by 3 rd party) used for mixing copper slag and air, ensure the availability and working of pressure relief valve and pressure gauge.	MEDIUM	

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

Fixing new Rubber (cold) sheets with adhesives.	Adhesive fumes	nose mask	LOW	
Fixing new rubber (Hot) sheet with steam application	Burning due to hot steam application • Adhesive fumes	Proper temperature to be ensured while entering for inspection during vulcanization of rubber.	MEDIUM	
Removing the scrap	Falling of scrap from height	Proper handling of scrap during handling	LOW	
Usage of Chemicals, adhesives, rubber solution etc. during rubber lining removal and relining	Fire Hazard	One Person to be available near manholes for emergency and to ensure the safe return of the persons inspecting/working inside the duct. • Fire Extinguisher to be kept ready near the vessel/tank. • Nearest Hydrant Point may be located, and hoses must be kept ready. • Usage of Non-Sparking Chisels and Tools is recommended. • Forced Ventilation is must during the entire activity.	HIGH	
		MSDS for the chemicals used must be displayed near the tank area. • Use of a lighter,		

TITLE	SAFETY PRECAUTIONS FOR HANDLING BULK ACID, CAUSTIC SODA and SODIUM CHLORITE
LMI No.	LMI/OGN/OPS/CHEM/007

		match box or any source of ignition must not be allowed within the vicinity. • Hot work is prohibited within the vicinity during rubber lining application		
PPE REQUIRED Helmet Safety Shoes Gloves Eye/Face Protection Hearing Protection Respiratory Protection Other PPE Other PPE		YES/NO		
Additional Precautions / Control Measures, if any	Full Body Harness at height work			
Further Precautions / Control Measures if any				
Toolbox Talk / Training to be given by				
This is to certify that all work under the scope of the above Job Plan shall be undertaken by qualified competent persons with experience of the type of work described above, and in all cases in full accordance with NTPC safety rules (E&M) and conditions of the safety document. What-If failure scenarios have been considered like failure of tools, tackles, work equipment in this JSA.				
Name of Area Engineer		Name of Head of Section		
Signature		Signature		
Declaration by the Agency: Our team has been clearly informed about safety aspects of this task. We have understood all points mentioned in the JSA and will follow these instructions in spirit. For any lapse in adherence to above safety instructions and consequences thereof, on behalf of our team working on this permit, I shall be solely responsible.				
Name and signature of site In charge/Supervisor				